

WHITE PAPER

Virtualizing the Infrastructure with Sun x86 Blades

Sponsored by: Sun Microsystems Inc.

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EXECUTIVE SUMMARY

Sun Microsystems has a long history of engineering systems that are optimized for network-enabled workloads. In the age of blade servers, this is translating into new designs for network enablement that improve the rate at which data is transmitted to and from the blades themselves. Sun has innovated by developing a sophisticated shared network interface card (NIC) called the Sun Blade 6000 Virtualized Multi-Fabric 10GbE Network Express Module (Sun Blade Virtualized NEM), which takes the place of embedded physical switches within the blade platform — and speeds the performance of the system, especially for enterprise applications accessing large data sets.

This white paper looks at ongoing customer requirements for server platforms and a number of pain points that have developed in recent years, as computer systems have become packed more densely within the datacenter. Power and cooling costs have risen at four times the growth rate of the actual costs of acquiring the servers themselves — and management costs have grown at eight times the acquisition-price growth rate. Beyond that, the sheer number of server footprints has multiplied so quickly that the overall solution has become too complex to manage easily or cost-effectively. Cabling for rack-optimized servers has also become tangled — figuratively and literally — meaning that any reduction in cabling would also improve operational efficiency.

IT staffers managing servers in the datacenter have also grown weary of configuring many types of servers over and over — and they are looking for ways to visualize the entire big-picture view of physical servers and logical servers with a single-pane-of-glass view that speeds remediation of any configuration or management issues. Reducing IT staff time associated with routine maintenance and management tasks is another way to make operations more efficient and less costly.

This IDC white paper describes new server blades from Sun Microsystems and a new shared NIC technology embedded within the Sun Blade Virtualized NEM, which are combined in a bladed server solution for the datacenter to address many of these operational issues. It also describes the competitive nature of the blade server segment, which continues to be one of the fastest-growing segments within the worldwide server market and has attracted the attention of some of the largest IT companies worldwide.

SITUATION ANALYSIS

Blade servers have been the focus of much attention in the datacenter in recent years, given the value proposition they provide: unified management of multiple blades; support for different types of processors and operating systems "under the same roof"; and the prospect of power/cooling cost reductions per physical server, which leads to improved energy efficiency for the enterprises that deploy blade servers.

This section of the white paper explores the dimensions of the worldwide blade server market, as it describes the ways in which blade servers address important, top-of-mind challenges for reducing ongoing costs within the datacenter.

Overview of Customer Requirements

The Cost of IT

In the continuing effort to support overall business demands, datacenter managers must overcome a distinct set of challenges. While it is still not completely clear what the recent economic slowdown will mean for IT spending during the next year, it is clear that IT organizations will be pressured to provide higher levels of service within increasing budgetary constraints.

Working against customers is the fact that the operational costs of running a datacenter have grown to eclipse the capital expenditure of the IT equipment. Due to a rapid explosion in the installed base of servers, datacenter budgets have been placed under duress by the personnel cost associated with managing and maintaining the server systems. In an effort to relieve some of this pressure, customers are focusing on solutions that effectively lower costs by simplifying their IT infrastructures and improving their management processes.

Over the past decade, two of the key factors that have driven the dramatic increase in the worldwide server installed base are a shift in the server mix and an expanding server footprint. Whereas once an organization would utilize only a handful of legacy mainframe or Unix systems to handle all of the IT load, today it is typical for a company to manage anywhere from several hundred to several thousand smaller servers distributed throughout the IT environment. To reduce the initial capital outlay, customers transitioned to the new, lower-priced scale-out technologies, most of which are based on x86 server technology. While the shift toward a distributed architecture allowed IT organizations to control new spending, a greater amount of effort (and expense) was necessary to manage the numerous systems.

Energy Expenses

The rising cost of energy combined with the trend toward green IT has placed considerable focus on the electricity expense necessary to run a datacenter; however, the impact of power and cooling on IT availability causes just as much if not more concern among IT managers. IDC surveys show that datacenter operators consider power and cooling their number 1 challenge today. IT organizations are constrained in their ability to deploy additional systems if the datacenter has reached the power and cooling thresholds. It is a common occurrence that the energy

envelope has become the primary limiting factor for operating budgets and the expansion of IT capacity. As part of their overall management strategy, datacenter managers are now seeking methods and solutions to control energy costs. By utilizing energy-efficient systems, IT managers are able to deploy more compute capacity in an existing datacenter with limited power, thus increasing productivity and alleviating the continuous pressure to expand the datacenter's footprint.

Consequently, there has been a shift in priorities over the past few years to a point where energy has become a central design point for the datacenter. In the past, the IT organization's objective was to maximize compute performance and expand IT availability and accessibility; the associated expense of power and cooling was largely an afterthought or simply viewed as a cost of doing business. The environment in the datacenter has changed, and datacenter customers are increasingly turning to solutions that can optimize server power and cooling. Relative to traditional rack-optimized servers, the integrated (shared chassis infrastructure) design of blades provides improved power efficiency. The pooling and sharing of the power distribution eliminates the need for multiple power supplies per server without sacrificing reliability.

IT Staff Costs, Time

IT staff costs are a major contributor to rising operational costs. These staff costs are related to maintenance, system administration, software repair and software upgrades, and training. To the extent that management tasks can be streamlined and simplified, and/or automated, the repetitive tasks that account for many IT staff hours can be reduced. In the case of blade server architecture, a unified management approach to all of the blades housed within a blade server chassis reduces the time that is traditionally taken up with monitoring and managing each individual server — as would be the case in rack-optimized server deployments. Further, the reduction in wiring and cabling between the servers and outside resources, supported by sharing of connections to the network and storage, reduces the deployment time and repairs of physical servers.

With the increasing adoption of server virtualization in the x86 server space, it is also vital to examine the increasing amount of system administrator time associated with tracking both physical machines and virtual machines (VMs) as they are deployed and provisioned with workloads. Indeed, the proliferation of VMs is expected to result in a "crossover" year in 2009, when the total number of VMs shipped outstrips the number of physical server units shipped worldwide. IDC notes that 2008 was a record year for physical server unit shipments, with more than 8 million units shipped worldwide.

Avoidance of Downtime

Downtime, whether caused by power outages, electrical outages, network outages, or human error, causes interruptions to business processes that "ripple" throughout the enterprise, affecting hundreds or thousands of end users and end customers. This is precisely why downtime is to be avoided, both by means of hardware redundancy and by avoiding single points of failure in system and software design.

High availability is an IT technology that avoids noticeable downtime, both by preventing interruptions to processing and by allowing work to be moved to alternate computing resources after a short interval for failover and restart. Redundancy in hardware design provides hot-pluggable components that can take over for each other, if need be, while high-availability software coordinates processing, even in the event that a given hardware component goes offline.

Avoiding unplanned downtime is another way to reduce operational costs — and it attacks lengthy periods of maintenance and repair by IT staffers. However, providing for "planned downtime" — or periods of time when hardware repairs can be made or software updates applied — is a good way to continue processing on other devices, even as repairs are made. In the case of blade servers, applications can be reprovisioned, or moved, to other blades while the originating blade server is taken offline. In cases of unplanned downtime, use of high-availability software ensures business continuity, as work is moved to available computing resources. The high-availability software moves working applications to other blades and back, as needed.

Sun's Nehalem-Based Blades

Hardware Architecture

- ☒ **Intel Xeon Processor 5500 Series.** In 2008, Intel announced the Nehalem microarchitecture, which leverages 45nm process technology to power quad-core processors for next-generation Intel Xeon x86 servers. Nehalem is a family name, and Intel has announced a series of Xeon processors, the 5500 Series, that build on this new microarchitecture, each addressing a specific price/performance range. Even so, all Nehalem processors will bring a new approach to staging data into and out of the processors, using the QuickPath Interconnect (QPI) technology to link processors with other components on the same platform. The QPI, which provides a direct-connect approach, replaces the longtime front-side bus (FSB) design of Intel Xeon processors, reducing potential I/O path contention.
- ☒ **Blade design.** The Sun blade platform leverages an open design to enable customers to select a configuration among multiple options that best suits their environment. Because Sun offers a choice in processor, operating system, and I/O, customers are able to avoid the "lock ins" of proprietary solutions. In this new launch, Sun is introducing a set of new x86-based server blades that complement the UltraSPARC-based blades housed within the same blade server chassis. This new generation of industry blades specifically targets virtualized environments by offering the maximum memory and I/O capacity. The new Intel blade, Sun Blade X6270 server module delivers a balanced end-to-end virtualization mechanism that maximizes the performance of the latest Intel CPU at an optimized cost point. This new blade features a two-socket quad-core blade that is capable of supporting 144GB system memory through 18 DIMM slots and provides 270Gbps of I/O through four PCI Express 2.0 x8 links, four SAS interfaces, and two Gigabit Ethernet interfaces. It complements the four-socket Intel-based 6-core blade with 24 FB-DIMM slots and the two AMD-based quad-core blades in the portfolio: One is a two-socket blade with 16 DDR2 DIMM slots providing 64GB of memory capacity, and the other is a four-socket AMD-based blade with 32 DDR2 DIMM slots supporting up to 256GB of memory and

two UltraSPARC blades. Additionally, the recently announced Sun Blade T6340 server module is capable of running 128 threads and supporting 256GB of memory via 32 LV FB-DIMM slots. These systems are able to provide ample memory, which allows the processing cores to support a higher number of VMs per physical server. For example, a fully configured Sun Blade 6048 chassis with the Sun Blade T6340 server module deployed with Sun's Logical Domain technology can handle potentially over 6,000 virtual domains. This can dramatically improve a customer's total cost of ownership by increasing system utilization, lowering hardware cost, and improving energy efficiency.

Customer Choice

- ☒ **Hardware choice in Sun blade chassis.** Blades based on Sun UltraSPARC, Intel Xeon, and AMD Opteron processors can be housed in the same blade server chassis. Sun has long been identified with its Solaris/SPARC systems; however, it is also true that the Sun Blade 6000 chassis houses a "mix" of server blades, including those based on SPARC, Intel, and AMD processors. This capability allows customers to choose the mix of hardware blades that is right for their datacenter, hosting the array of applications that they are already using. IDC worldwide server research shows that the most widely used operating environments, including Unix, Windows, and Linux, are supported and are employed by customers aboard Sun Blade 6000 server modules. In addition, hypervisors such as VMware and Sun xVM are supported natively on Sun Blade 6000 server modules. These hypervisors can host Unix, Windows, or Linux guest operating systems.
- ☒ **Software choice on x86 server blades.** Sun x86 server blades support choice in operating systems (Solaris; OpenSolaris; Microsoft Windows; and multiple Linux distributions, including Novell SUSE, Red Hat RHEL, and others) and in hypervisors for hosting virtualized workloads. These hypervisors include VMware ESX Server, Microsoft Hyper-V, and Sun xVM Server. In turn, the hypervisors support virtualized "guest" operating systems, including Microsoft Windows, Linux, and Solaris. Other virtualization technologies include Sun's Virtual Box/VDI, for supporting virtualized applications, and Solaris containers, which are software-defined partitions within a single Solaris domain that allow applications to run within protected and isolated virtual spaces. The significance of this broad support is that it allows businesses to deploy a full complement of workloads (e.g., business processing, database; decision support, collaborative email; high-performance computing [HPC]; IT infrastructure and Web infrastructure); as well as a deep inventory of independent software vendor (ISV) (packaged software) and custom applications that were specifically developed for vertical markets or for individual enterprises. For purposes of supporting cloud computing, the strong support for virtualization on Sun x86 blade servers means that cloud services can be provided to end users, using Sun blades for hardware components.

The Sun Blade 6000 Chassis

In 2007, Sun introduced the Sun Blade 6000, the enterprise-class version of its modular platform. The modular architecture of the blade platform delivers flexibility and agility to a customer's IT infrastructure. The consolidation of multiple server systems into a single chassis with shared power, cooling, and I/O improves the

overall efficiency and reliability of the IT systems. Additionally, the hot-swappable feature of the form factor improves the serviceability aspect.

The Sun Blade 6000 was designed to be an open platform providing a greater amount of choice and flexibility for today's datacenters. Customers are able to deploy three processor architectures — Sun UltraSPARC, AMD Opteron, and Intel Xeon — in the same chassis. In addition to the choice of processors, the Sun Blade 6000 can support Solaris, Linux, and Windows operating systems and the VMware ESX Server virtualization environment, enabling customers to handle a variety of workloads, from low-end infrastructure applications to higher-end mission-critical applications.

Sun has stressed the simplicity in the design and architecture. Specific features include the following:

- ☒ **Transparent management.** The Sun Blade 6000 utilizes the same management module found in Sun rackmount servers. Customers are able to manage their infrastructure across blade and rackmount platforms through a single management module.
- ☒ **Independent I/O.** Sun opted for an industry-standard I/O design that leverages PCI Express (PCIe) in lieu of a proprietary solution. The open architecture design of the chassis midplane was developed to lower cost and improve flexibility as more PCIe choices become available to customers.
- ☒ **Sun "No-Compromise Cooling" features.** An enhanced air-flow design improves the energy efficiency of this blade design, thereby greatly reducing costs associated with power and cooling. Leveraging the shared power and cooling infrastructure of the chassis, Sun's blades benefit from the use of fewer, yet larger and more efficient power supplies and cooling fans. Each individual rack server requires its own power supplies and fans, which is a less efficient design than the shared power and cooling design of the blade chassis.
- ☒ **Virtualization ready.** Even though virtualization had yet to achieve mainstream adoption in late 2006, Sun designed the Sun Blade 6000 with the idea that virtualization would be widely deployed in customer sites using x64 technologies from Sun. First, Sun developed a relatively larger blade than those common in the industry to provide ample space for the additional memory required by virtualized systems. Second, the system was designed with a higher I/O throughput to ensure the I/O would not be the bottleneck for the virtualized environment. Some of the benefits of Sun's large blade design are that these blades have the physical real estate required to provide large amounts of memory (more DIMM slots) and very high I/O bandwidth, making the Sun blades optimized for deploying virtual environments. Additionally, by not having to compromise on memory or I/O, Sun's customers are able to maximize ROI by meeting today's IT SLAs while ensuring that they have the necessary headroom for the future.

Software Solutions

Although Sun is best known for its products that were developed in the Unix/RISC server market space, its x86 servers support a wide variety of operating environments. In addition to Solaris, which is Sun's form of the Unix operating system, Sun's x86

servers support Microsoft Windows, a number of Linux distributions (e.g., Novell SUSE and Red Hat RHEL, among others), as well as VMware ESX Server and Microsoft Hyper-V virtualized computing environments. This means that Sun x86 server blades support customer choice in software environment and in terms of the range of applications that can run on the server blades. Support for this diversity in software environments reflects the adoption of many types of applications within the enterprise — and a unified management approach makes it easier now to run all of these applications on a single piece of hardware equipment.

A range of software solutions runs on top of these operating systems, including middleware for transaction processing, support of Java applications and .NET applications, as well as provisioning and management capabilities. Sun's system software is leveraged, including the Java language and open source application development environment, both OpenSolaris and Solaris, the MySQL open source database, and Glassfish for support of software appliances. Sun supports a deep inventory of software from ISVs and works with its channel partners to distribute software solutions to end customers.

Solaris Optimization from Sun-Intel Alliance

In January 2007, Sun and Intel announced an alliance centered on an agreement between the two companies to optimize Solaris performance on Xeon processors. At that time, Intel said that it viewed the Solaris operating system as an enterprise-class, mission-critical Unix operating system. Engineers from both companies have worked together to optimize system design; this has improved both the performance of Solaris running on Intel Xeon systems and the reliability and power efficiency of these Solaris/Xeon servers. Intel is one of the top contributors to the open source OpenSolaris community, through which Sun and the community develop next-generation functionality that will become integrated with Solaris 10 over time. This process encourages innovative development through an open source model while allowing time for new features to be tested and certified for use in Solaris 10, which has long been used in IT deployments. In recent years, Sun has begun to supply support for OpenSolaris as well so that many companies have also begun to deploy both OpenSolaris and Solaris for production workloads.

Sun and Intel have also worked on power management functions, improving energy efficiency and performance per watt through Integrated Power management and the new Solaris Power Aware Dispatcher. Importantly, the Solaris Fault Management Architecture (FMA) infrastructure has been enhanced to take advantage of Intel processor reliability, availability, and serviceability (RAS) features so that transient processing errors can be isolated and will not cause production processing to stop.

Sun's Approach to the Marketplace

IT Benefits

Sun is supporting an open x86 server platform, providing customer choice in terms of the software environments that run on the x86 server blades. This type of choice allows IT organizations to have more flexibility in terms of software deployments — and it supports virtualized software "stacks" as they are provisioned across the x86 server infrastructure of customers' datacenters.

Although Sun was once identified as providing Solaris solutions only on its SPARC hardware platforms, now Solaris has been optimized to run on Intel Xeon Processor 5500 Series quad-core processors based on the Nehalem microarchitecture. The Nehalem processors support Intel Virtualization Technology (VT) features, enhanced memory features, and the new QPI links between processor components, which is the successor technology to Intel's FSB on-chip interconnects. QPI is expected to improve overall Intel x86 throughput, compared with earlier Xeon processors, as it reduces the path length to memory and improves on-chip I/O.

How Sun Is Addressing IT Challenges

Business Benefits

Sun's blade servers address many of the top causes of operational costs in today's datacenter, and by reducing these operational costs, they support ongoing business processes more efficiently. Examples of the types of costs addressed by the blade servers include energy efficiency in the blade server design, which reduces power/cooling costs; reduction in the amount of datacenter floor space that is needed to house the blade servers compared with rack-optimized servers; and costs related to IT staff time and training, which saves funds that can be used for other business purposes. IDC has often drawn the connection between IT infrastructure and business requirements and the need to align the two more closely in order to ensure business agility. Current economic conditions place a high priority on gaining efficiency in IT operations as businesses conserve financial resources and prepare to meet the challenges of the new business climate worldwide.

Sun Blade Virtualized NEM Technology

How It Works

Sun is introducing new I/O module technology for the Sun Blade 6000 chassis. The Sun Blade Virtualized NEM includes a "virtualized NIC" that removes a layer of physical switching within the chassis itself. What this means to IT managers is that fewer NICs need to be attached to each server module, allowing the design itself to be less complex and reducing overall power requirements as well. The design goal is to improve performance and I/O throughput, especially impacting heavily virtualized workloads.

As the adoption of virtualization continues to rise — often in conjunction with workload consolidation projects — improvements in I/O performance will also become increasingly important. Without it, the intensive use of hypervisor software, supporting higher numbers of VMs per physical server, would likely slow overall performance for demanding enterprise applications.

Sun examined this operational issue and developed the new NEM with virtual I/O technology that permits up to 10 server modules to share a common high-performance 10GbE NIC. For more bandwidth-intensive applications, the Sun Blade Virtualized NEM can be partitioned to utilize its two shared 10GbE NICs, each of which is bound to five server modules. In both configurations, constituent modules can also communicate to one another at 10GbE rates. Additionally, when the

Sun Blade Virtualized NEM is partitioned, leveraging both 10GbE uplinks, the NEM protects live application sessions (across all 10 blades) in the event that a link is lost.

Further, the Sun Blade Virtualized NEM provides four Sun registered MAC addresses to each server module for binding to the VM's virtual NIC (VNIC), resulting in 40 VNICs per Sun Blade Virtualized NEM or 80 per chassis using two Sun Blade Virtualized NEMs. Perhaps the greatest benefit of using the Sun Blade Virtualized NEM, beyond a 10:1 reduction in cables, is its intelligence and automation, which removes the overhead that would be associated with in-chassis switching.

IT Benefits

Customers have realized the benefits of consolidating their disparate IT workloads into the unified blade platform. The blade segment has been one of the fastest-growing segments; IDC estimates that 13.2% of all servers shipped worldwide in 2008 were blade servers. Given the new designs and enhancements vendors are bringing to market, IDC expects this trend to continue to a point where blades will account for 29% of all servers by 2013.

Virtualization adoption on x86 servers is increasing, with the average use of two to four VMs per physical server just a few short years ago rising to eight, or more, VMs per physical server at many sites. The increasing VM density, combined with the presence of important enterprise applications running inside the VMs, means that the amount of system bandwidth to support all those workloads must increase as well. One of the ways to address this is to virtualize the I/O for all of the connections into the servers, or the blade servers, running in the datacenter.

Blade and virtualization technologies have always been complementary, and Sun is further building upon this synergy with the virtualized I/O of the Sun Blade Virtualized NEM. The challenge of virtualizing I/O has become more pressing in 2008 and 2009 — as IDC estimates that more than 30% of x86 server infrastructure is now running as virtualized IT infrastructure — headed for 50% or more by 2010. And yet, many I/O and memory "bottlenecks" have prevented this dense computing environment from reaching its full potential for hosting the most demanding enterprise applications and database workloads. Sun's design work with its Sun Blade Virtualized NEM shared NIC capability, which leverages built-in hardware virtualization technology, has virtualized the NICs that are housed on every blade server in the Sun Blade 6000 chassis. This optimizes server I/O while allowing more blades to be housed within a relatively compact blade package.

Business Benefits

The mounting costs associated with IT facilities and the continuing surge in the number of physical devices that are being deployed are impacting the efficient operation of the datacenter. While the real estate constraints have many customers considering additional buildout or major retrofits, other sites that may have ample floor space are challenged with power and cooling issues.

As companies look to move forward in this next IT era, the idea of reducing the amount of datacenter floor space needed to support the business while maximizing the amount of compute power that can be housed within that space poses a significant IT challenge

for enterprises and service providers. Technologies that help businesses achieve these goals, as they increase computing capacity and IT flexibility, are allowing the datacenter to rapidly adapt to changing business conditions.

To allow businesses to "do more with less," all existing workloads must be made to fit within existing datacenter thermal envelopes and even a company's financial envelopes imposed by today's challenging business conditions. The ability to host a range of enterprise workloads — applications and databases included — within the Sun Blade 6000 chassis and to move workloads around this bladed infrastructure, as needed, with ample I/O capacity gives businesses a new way to consolidate their current IT inventory and to contain ongoing maintenance costs.

The traditional one-off approach to the datacenter, with one application running on each server, has created a number of challenges for IT organizations, and these hurdles have become particularly evident recently as existing datacenters start to age. Power and cooling issues are at the forefront of many datacenter concerns, and retrofitting existing datacenters to account for the increasing density of servers and storage devices has become something of a science project for many customers as there is no simple solution for the more than 10,000 individual enterprise-class datacenters worldwide. The list of solutions and "best practices" for solving today's datacenter challenges is long and varied, which is typical of any emerging market in which solutions quickly become fractured due to lack of standards.

Support and Services

Sun Support Services

All Sun blades have a three-year standard hardware warranty, with next-day business response as a standard level of service. There is also a chassis-based service option, which is an upgrade from the standard warranty, that provides coverage for the quantity and types of blades and infrastructure modules within the chassis — and for the chassis itself.

As for all types of Sun servers, Sun also offers Sun Spectrum services, in which customers pay for the level of services required — Silver, Gold, and Platinum — depending on the frequency of service and the response times expected. In addition, Sun offers installation services as well as quick-start and application readiness services.

Sun Microsystems Global Financial Services also offers a Technology Refresh service, in which customers can easily acquire technology when business needs dictate, without the restrictions of financial budgets. This service offering gives customers the freedom and flexibility to upgrade or add to their blade infrastructure at any time, without the worry of technology obsolescence.

Sun's Customer Ready System offering provides preconfigured blade installations, which are assembled in and provisioned from the factory before being delivered to customers.

Sun Professional Services

Sun offers professional services aimed at architecting and implementing blade server deployments, migrating workloads, assessing the current IT infrastructure, and helping customers to manage the bladed server infrastructure, after it is deployed. Because workloads are often mapped differently to run on blade servers, compared with rack-optimized servers, professional services can aid the assessment and implementation processes, leveraging experience gained over many customer engagements.

Sun's architectural services include consolidation and virtualization services; datacenter strategy, design, and build services; and environmental services. Implementation and migration services include application readiness services, data migration services, and a Sun Express Service aimed at HPC. Assessment services offer performance analysis and capacity planning services and power/cooling and energy-efficiency services. Examples of onsite management services include onsite as well as remote operations management services.

Challenges, Meeting the Challenges, and Opportunities

The blade server market is one of the fastest-growing segments of the worldwide server market, but it is also one of the most competitive. A relatively small number of top vendors compete with each other, based on performance, price/performance, performance/watt, and management of all blades within the chassis. Considerations about energy efficiency are also important because densely packed blades must be designed to emit as little heat as possible, given the design constraints of blade server architectures. Further, the need to help customers plan how workloads will be provisioned across the blade servers is another customer requirement that all blade vendors must address.

Sun, a longtime provider of densely packed rack-optimized servers, entered the blade market in 2006, applying its engineering resources to the task of designing blade chassis and x86 blades (which Sun terms x64 blades). Sun's design center is leveraging its experience in producing energy-efficient servers, with high performance-per-watt ratios, to keep the "thermal envelope" of its blade servers low. Its management software, including ILOM remote management for lights-out datacenters, is available for its x86 blade servers, as is its provisioning software under the Sun xVM management framework. Importantly, Sun is addressing the full range of operating systems and hypervisors available for the x86 server marketplace. These capabilities position Sun to meet the IT and business requirements of enterprises that are considering deployments of x86 blade server infrastructure.

CONCLUSION

Blade servers are seeing rapid adoption in the x86 server marketplace due to their performance and management characteristics. They give IT greater flexibility in deploying new workloads — and in reallocating workloads to available compute resources as business needs change over time. Blades are often

associated with virtualization and workload consolidation projects, as businesses set about the task of transforming the datacenter to support improved operational characteristics.

Sun Microsystems blades are optimized for network-enabled workloads. The new Sun Blade Virtualized NEM supports I/O virtualization, improving the amount of I/O available to virtualized server blades running within the Sun blade systems. This change in architecture supports the increasing adoption of virtualization on x86 infrastructure — and the migration of enterprise workloads, which require high levels of I/O, to blade server systems. By designing and implementing these blade servers, Sun is providing a platform for highly virtualized workloads, for flexible IT, and for improved operational characteristics, all of which reduce ongoing operational costs, which benefits business.

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